

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA0304

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).

Assessment Tool Weightage: 25%

Dr DUMPALA PRASANTH

QUESTIONS: 10

Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

(5 Marks)

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

(5 Marks)

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm

converges faster based on the observed results.
(5 Marks)

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. **(5 Marks)**

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. **(5 Marks)**

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability. **(5 Marks)**

7. A Q-learning agent is tested with different exploration strategies: ϵ -greedy, softmax, and random exploration. The cumulative reward is highest with ϵ -greedy. Compare the strategies and defend why ϵ -greedy achieves better performance in this scenario. **(5 Marks)**

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy. **(5 Marks)**

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective. **(5 Marks)**

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning. **(5 Marks)**

Code of Conduct:

I (K Varshan / 192325089) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K Varshan

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

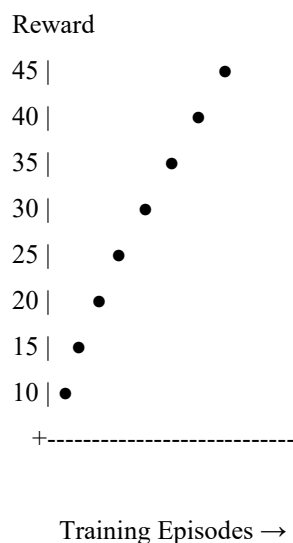
Reinforcement Learning – Data Interpretation Answers

1. Reward increases from 10 to 45

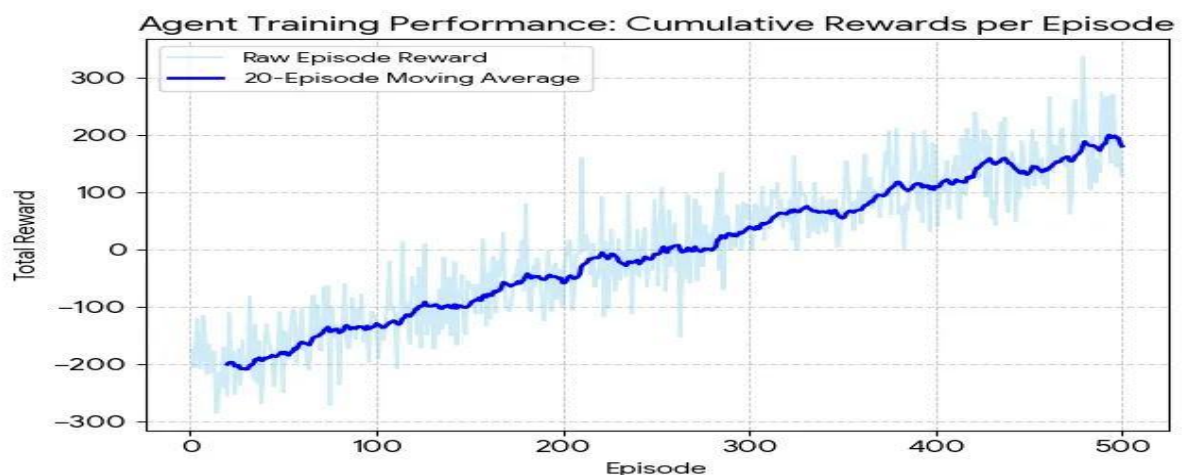
The reward increases gradually from **10 to 45** over multiple training episodes.

Interpretation

Reward represents the feedback received by the agent for its actions. An increasing reward indicates that the agent is learning from its previous experiences and improving its decision-making strategy.



Diagram



The gradual increase suggests:

The agent is learning useful state-action relationships.

The policy is improving with experience.

The agent is selecting better actions.

The cumulative performance is increasing.

Exploration is gradually producing better knowledge.

Since the reward reaches **45**, the agent is performing considerably better than at the beginning.

Conclusion

The increasing reward trend is a positive indication of **learning and policy improvement**. However, reaching an optimal policy cannot be confirmed only from increasing rewards; the reward should ideally stabilize near a high value over many episodes.

2. Epsilon values 0.1, 0.3 and 0.5

The epsilon-greedy strategy controls the balance between **exploration and exploitation**.

$\varepsilon = 0.1 \rightarrow 10\%$ exploration, 90% exploitation

$\varepsilon = 0.3 \rightarrow 30\%$ exploration, 70% exploitation

$\varepsilon = 0.5 \rightarrow 50\%$ exploration, 50% exploitation

Given that the highest cumulative reward occurs at $\varepsilon = \mathbf{0.1}$, the agent performs better when it exploits its learned knowledge more frequently.

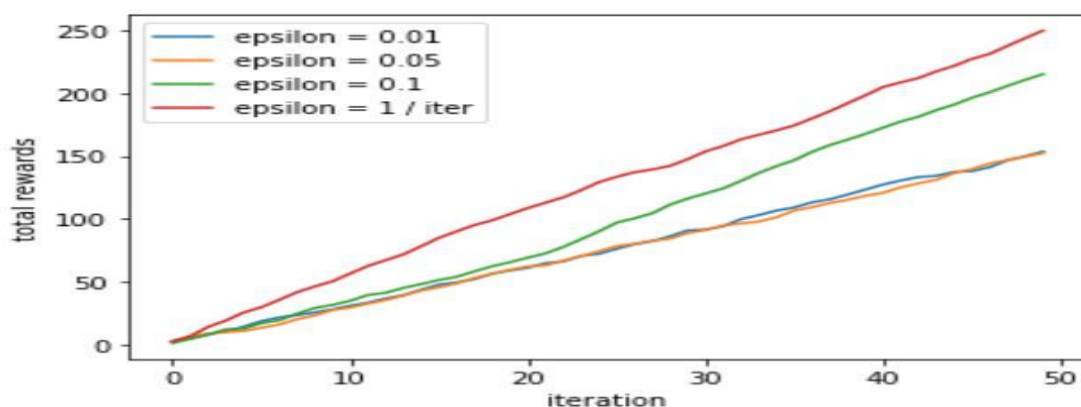
$\varepsilon = 0.1 \rightarrow$ Mostly exploit $\rightarrow$ Highest reward

$\varepsilon = 0.3 \rightarrow$ Balanced $\rightarrow$ Medium reward

$\varepsilon = 0.5 \rightarrow$ More explore $\rightarrow$ Lowest reward

At $\varepsilon = 0.5$, the agent chooses random actions more frequently. These unnecessary exploratory actions can reduce the immediate cumulative reward.

Diagram



Conclusion

For this particular environment, $\epsilon = 0.1$ provides the best performance because the agent has sufficient knowledge to exploit good actions. A lower epsilon does not always guarantee better performance, but the given results clearly show that 0.1 is optimal among the tested values.

3. Q-Learning vs SARSA

The two algorithms are compared using average rewards over **100, 200, and 300 episodes**.

The observation is:

Q-Learning → Slightly higher rewards

SARSA → Slightly lower rewards

Analysis

Q-Learning is an off-policy algorithm. It updates its Q-value using the maximum estimated value of the next state:

$$[Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]]$$

SARSA is an on-policy algorithm:

$$[Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma Q(s',a') - Q(s,a)]]$$

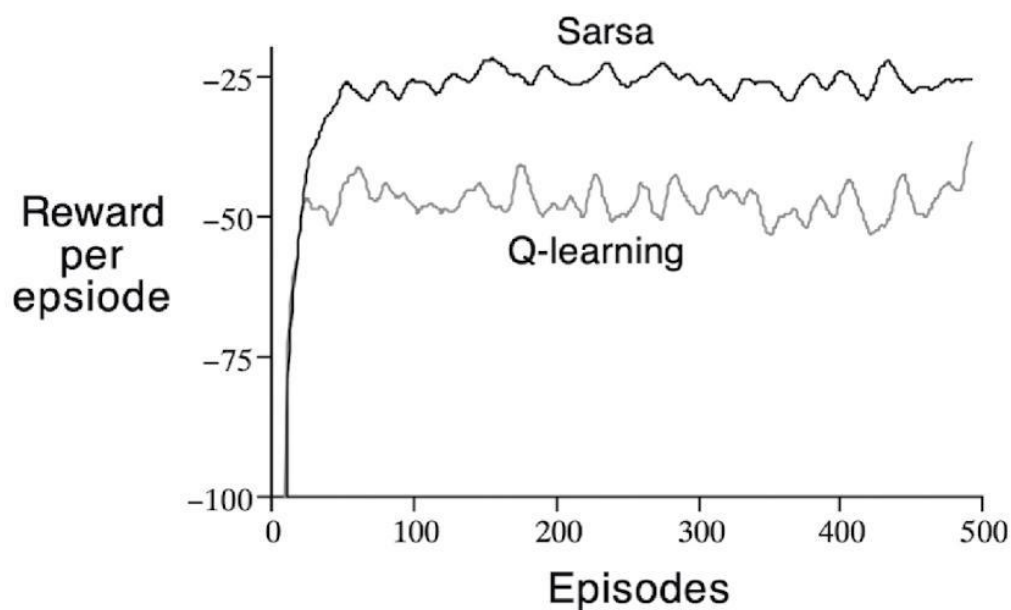
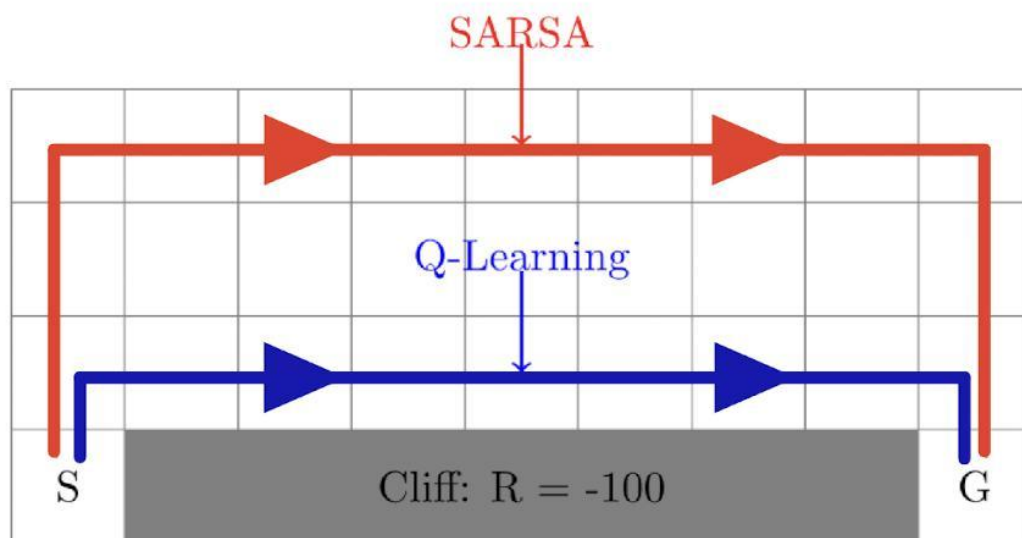
Therefore, Q-Learning directly learns toward the best estimated future action, whereas SARSA learns according to the action actually selected by its current policy.

Interpretation

If Q-Learning consistently produces higher rewards at 100, 200 and 300 episodes, it demonstrates better performance under the given experimental conditions.

However, **higher reward alone does not prove faster convergence**. To determine convergence speed accurately, we should observe when each algorithm reaches and maintains a stable reward.

Diagram



Conclusion

Based on the stated results, **Q-Learning performs better** because it consistently achieves slightly higher rewards. If it reaches a stable high reward earlier than SARSA, it can also be considered faster-converging.

4. Discount Factors 0.2, 0.5 and 0.9

The discount factor (γ) determines how strongly the agent considers future rewards.

$$[0 \leq \gamma \leq 1]$$

The tested values are:

$\gamma = 0.2 \rightarrow$ Strong focus on immediate rewards

$\gamma = 0.5 \rightarrow$ Moderate consideration of future rewards

$\gamma = 0.9 \rightarrow$ Strong focus on future rewards

The average reward increases as γ increases.

The return is:

$$[G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots]$$

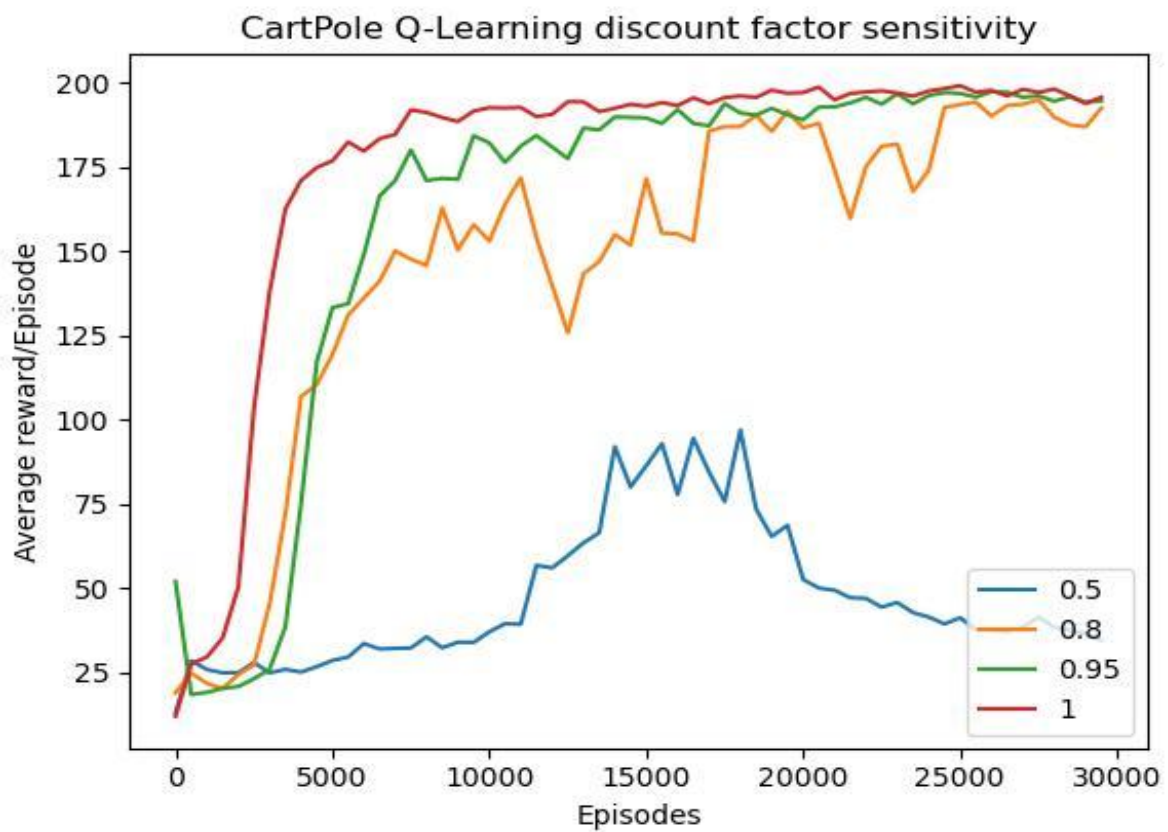
At $\gamma = 0.2$, future rewards are heavily discounted.

At $\gamma = 0.9$, future rewards retain much more importance.

Interpretation

The increasing reward indicates that the environment benefits from **long-term planning**. The agent performs better when it considers future consequences rather than only immediate rewards.

Diagram



Conclusion

Among the tested values, $\gamma = 0.9$ is the most suitable because it produces the highest average reward. It enables the agent to make decisions that provide greater long-term benefits.

5. Success Rate increases from 45% to 92%

The success rate increases from:

[
45%\rightarrow92%
]

This represents an improvement of:

[
 $92-45=47\text{ percentage points}$
]

The relative improvement is approximately:

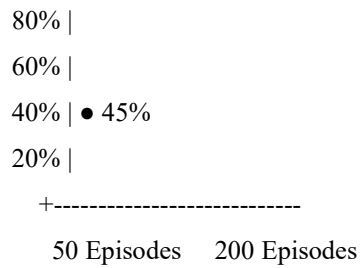
[
 $\frac{92-45}{45}\times100$
 $\approx104.4\%$
]

Interpretation

The significant increase indicates that the agent is learning from its interactions with the environment.

Success Rate

100% | ● 92%



Possible reasons for improvement include:

- Better state-action knowledge.

- Improved policy.

- Reduced selection of poor actions.

- Better understanding of the environment.

- More effective exploitation of learned information.

Does 92% prove an optimal policy?

Not necessarily.

A **92% success rate strongly indicates effective learning**, but optimality requires additional evidence such as:

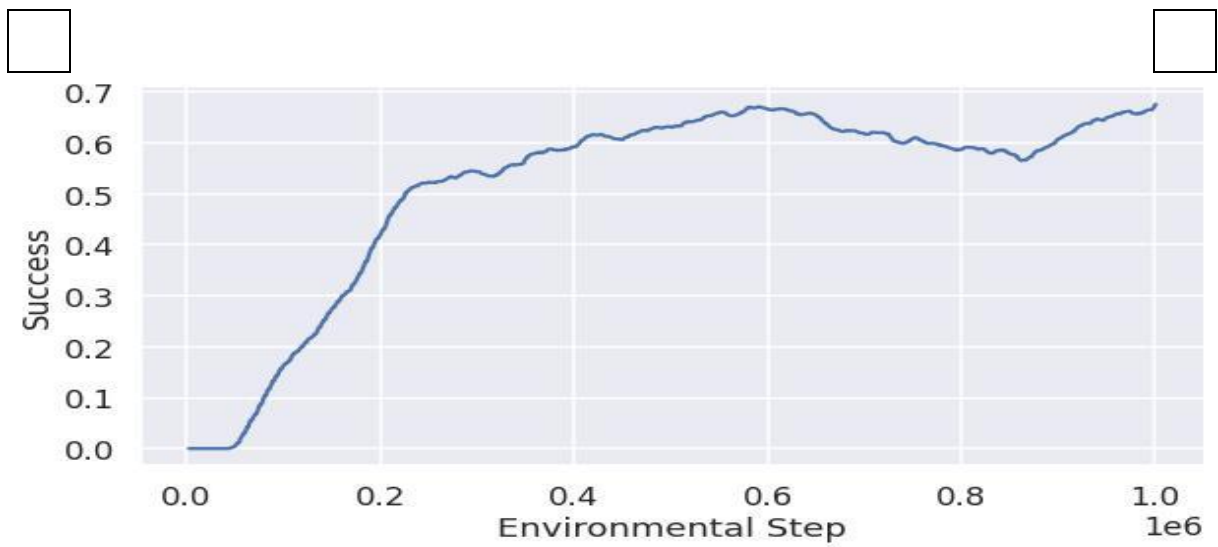
- Stable performance over further episodes.

- Comparison with the theoretical optimum.

- Low variance.

- No significant improvement with additional training.

Diagram



Conclusion

The increase from **45% to 92%** strongly indicates substantial policy improvement, but the data alone is insufficient to claim that the policy is fully optimal.

6. Learning Rates 0.01, 0.05 and 0.1

The learning rate controls the size of parameter updates during training.

The tested values are:

0.01 → Small updates

0.05 → Moderate updates

0.10 → Large updates

Effect of Learning Rate

A very small learning rate such as **0.01** can make learning slow because the model changes its parameters only slightly after each experience.

A very high learning rate such as **0.1** can cause unstable learning because the updates may be too large.

The value **0.05** provides a middle ground.

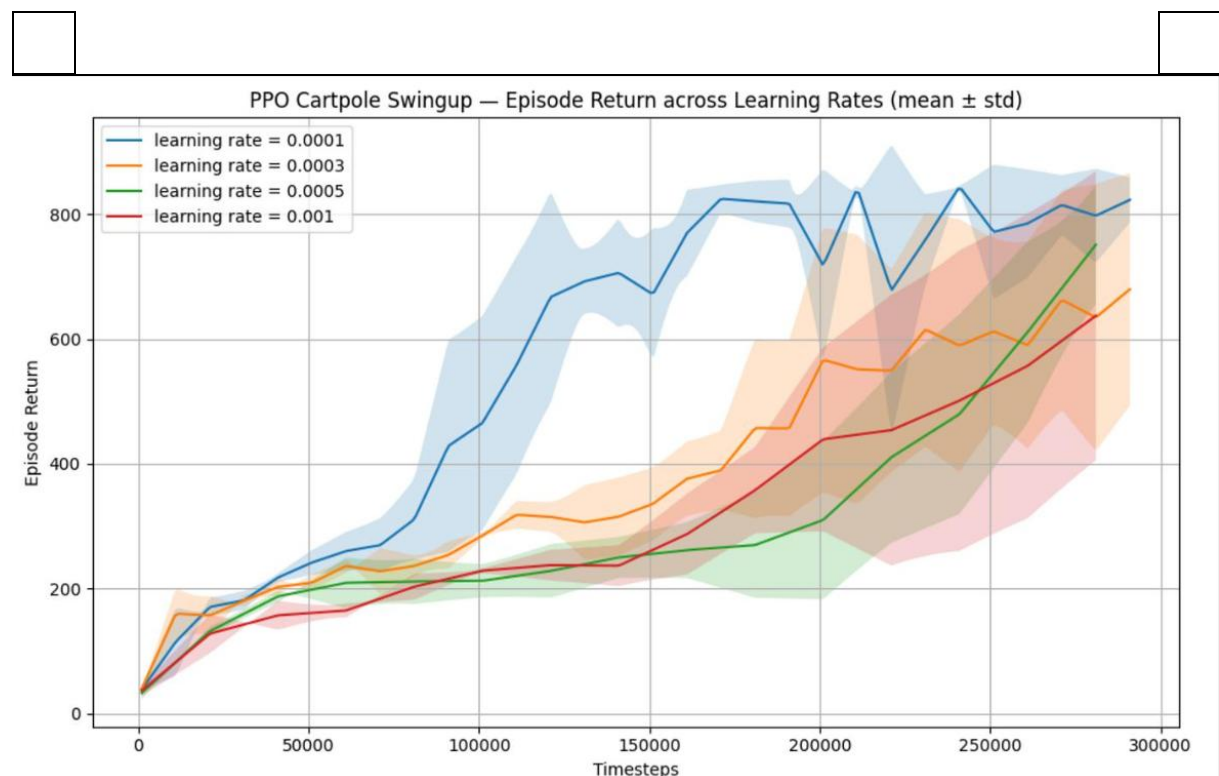
Learning Rate

|
Slow | 0.01
|
Best | 0.05 ← Stability + reasonable speed
|
Risk | 0.10
|

Interpretation

Since the agent achieves the **highest stability at 0.05**, this indicates that the update size is appropriate for the CartPole environment.

Diagram



Conclusion

Learning rate = 0.05 provides the best trade-off between learning speed and stability among the tested values. It avoids the slow convergence of 0.01 and the possible instability of 0.1.

7. ϵ -Greedy vs Softmax vs Random Exploration

The three exploration strategies are:

ϵ -Greedy

Chooses the best-known action most of the time and randomly explores with probability ϵ .

Best action $\rightarrow$ Exploit

Random action $\rightarrow$ Explore

Softmax

Selects actions according to their estimated Q-values.

Higher-valued actions receive higher probabilities.

$$P(a|s) = \frac{e^{Q(s,a)/\tau}}{\sum_b e^{Q(s,b)/\tau}}$$

Random Exploration

Selects actions randomly without strongly considering learned Q-values.

Comparison

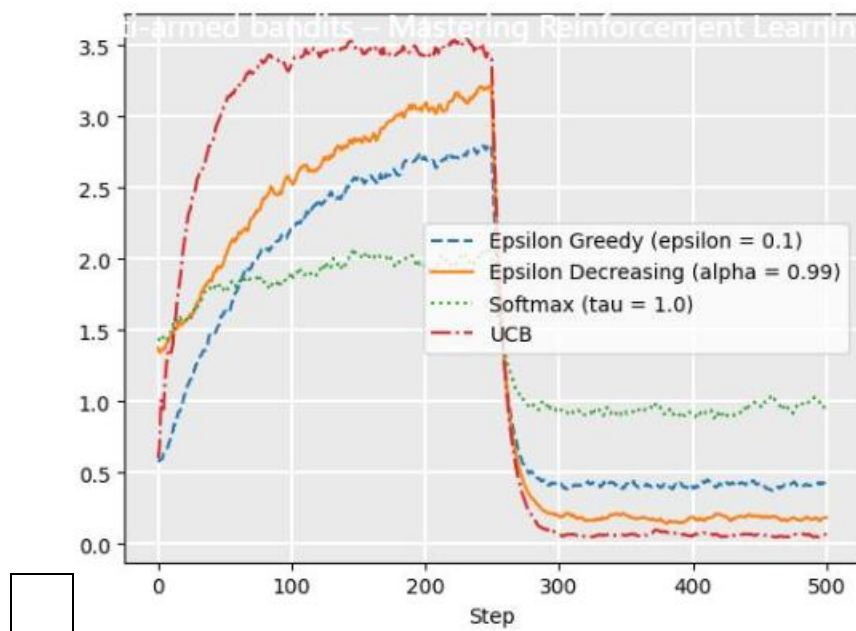
Strategy	Decision Method	Expected Behavior
ϵ -Greedy	Mostly best action	Strong exploitation
Softmax	Probability based on Q-values	Balanced exploration
Random	Completely/randomly explores	High exploration

Since ϵ -greedy produces the **highest cumulative reward**, it is the most effective strategy in this particular experiment.

Reason

The environment apparently benefits from exploiting already learned good actions. Random exploration wastes more actions, while softmax may continue exploring actions that have lower estimated values.

Diagram



Conclusion

ϵ -greedy performs best under the given conditions because it provides a simple and effective balance between exploration and exploitation while allowing the agent to exploit successful actions frequently.

8. Maze: Steps decrease from 50 to 15

Initially, the agent requires:

[
50\text{ steps}
]

After 300 episodes:

[
15\text{ steps}
]

The reduction is:

[
 $50 - 15 = 35$ \text{ steps}
]

Percentage reduction:

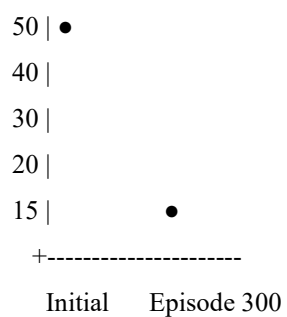
[
 $\frac{35}{50} \times 100 = 70\%$
]

Therefore, the number of steps decreases by **70%**.

Interpretation

The agent has learned a more efficient path to the goal.

Steps



This suggests:

Better knowledge of the maze.

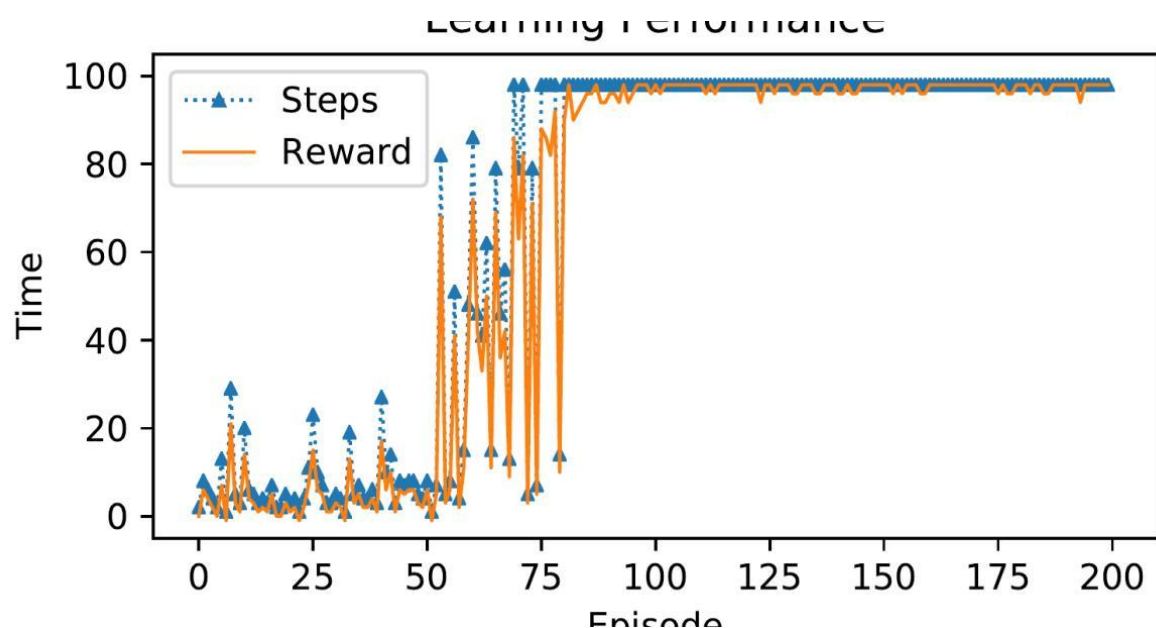
Fewer unnecessary movements.

Improved action selection.

More efficient policy.

Better exploitation of learned information.

Diagram



Does it prove optimal convergence?

Not completely.

If the shortest possible path is **15 steps** and the agent consistently achieves 15 steps, then this would strongly indicate convergence to an optimal policy.

If the actual shortest path is unknown, the reduction only demonstrates **policy improvement**, not guaranteed optimality.

Conclusion

The 70% reduction in steps indicates significant learning. Further stable results are needed to confirm convergence to the optimal policy.

9. SARSA with $\gamma = 0.4, 0.7$ and 0.95

The discount factor determines the importance assigned to future rewards.

$\gamma = 0.4 \rightarrow$ Mostly immediate rewards

$\gamma = 0.7 \rightarrow$ Moderate future consideration

$\gamma = 0.95 \rightarrow$ Strong long-term consideration

The return is:

$$\begin{aligned} &[\\ G_t &= R_{t+1} + \gamma R_{t+2} \\ &+ \gamma^2 R_{t+3} + \dots \\ &] \end{aligned}$$

At $\gamma = 0.95$, future rewards are discounted only slightly.

Interpretation

The highest long-term reward occurs at $\gamma = \mathbf{0.95}$. This indicates that the SARSA agent benefits from considering future consequences strongly.

For example, an action that gives:

Immediate reward = -2

Future reward = +20

may be preferable when γ is high because the future benefit is given significant importance.

Comparison

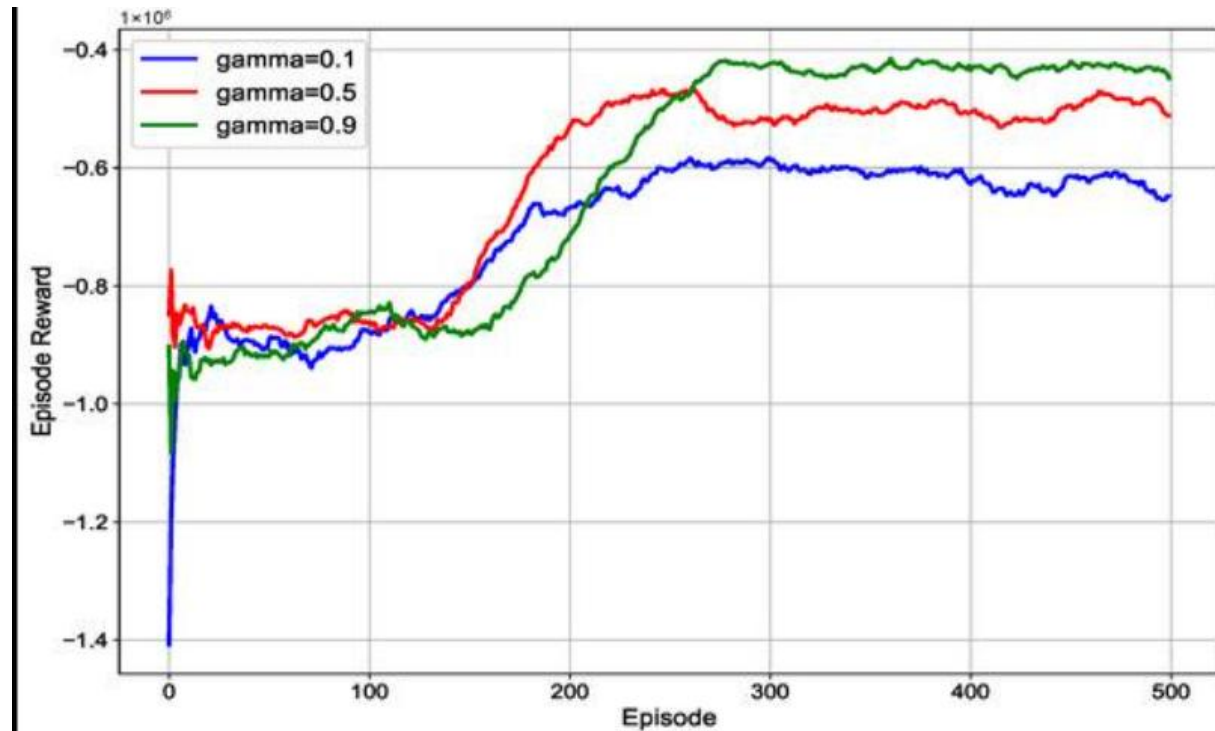
γ **Decision Preference**

0.4 Short-term

0.7 Balanced

0.95 Long-term

Diagram



Conclusion

$\gamma = 0.95$ is the most effective value in this experiment because it produces the highest long-term reward. The environment appears to require long-term planning rather than purely immediate decisions.

10. Fluctuating Rewards Stabilize After 500 Episodes

The agent experiences fluctuating rewards during the early training episodes but becomes stable after approximately **500 episodes**.

Early Training

During early episodes, the agent has limited knowledge and performs significant exploration.

Therefore:

Early episodes



More exploration



Different actions



Fluctuating rewards

Later Training

After approximately 500 episodes:

More experience



Better Q-values / policy



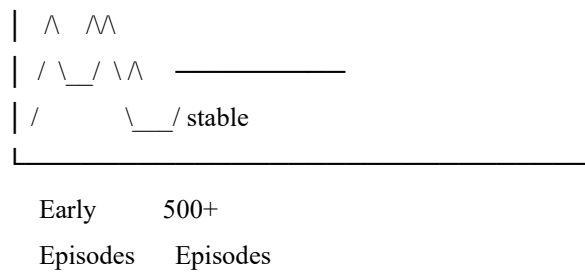
Better action selection



Stable rewards

Reward Trend

Reward



The stabilization indicates that the agent has learned a relatively consistent strategy.

Policy Improvement

Stable rewards suggest:

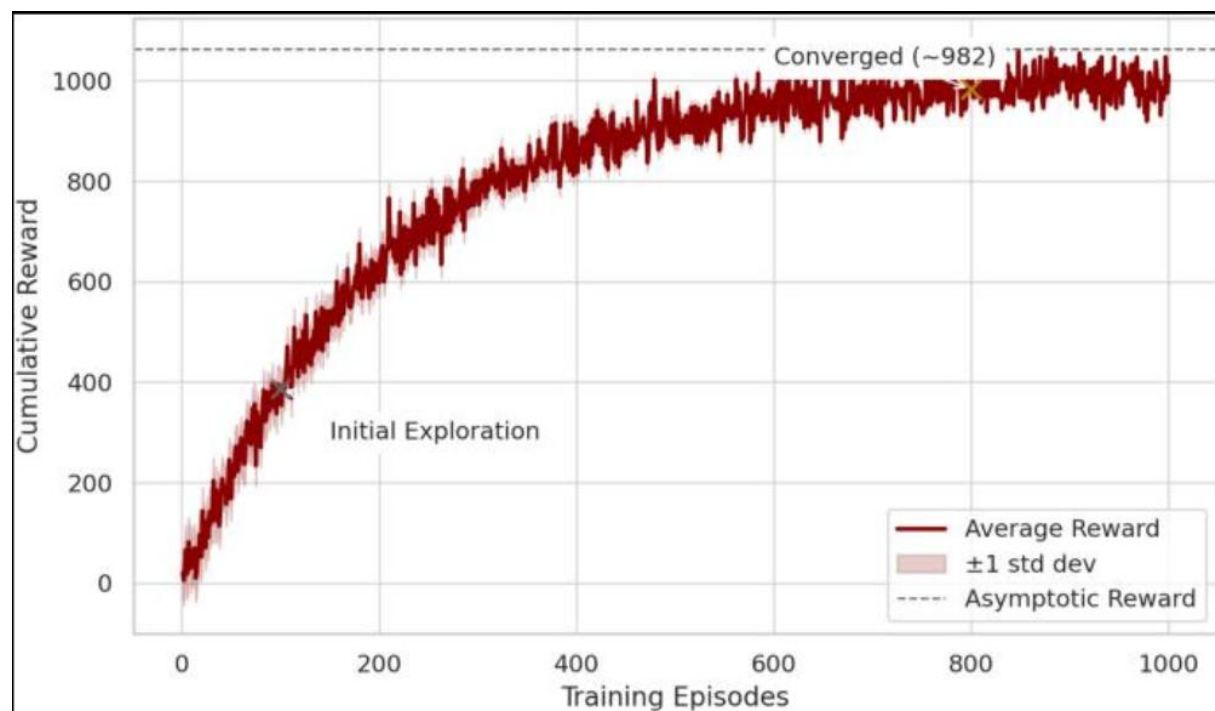
The agent has learned useful state-action relationships.

Random exploration has less influence on performance.

The policy is becoming consistent.

The agent is repeatedly achieving similar outcomes.

Diagram



Robustness

In a stochastic environment, some variation may always remain because the environment itself contains randomness. Therefore, complete elimination of fluctuations is not necessary.

Conclusion

The stabilization of rewards after 500 episodes provides evidence of **policy improvement, learning convergence, and increased robustness**. However, stability should ideally be confirmed using average reward, variance, and performance over additional unseen episodes.

Final Revision Table

Q	Given Observation	Main Interpretation	Best Conclusion
1	Reward $10 \rightarrow 45$	Learning improves	Policy is improving
2	$\epsilon=0.1$ gives highest reward	More exploitation works better	$\epsilon=0.1$
3	Q-Learning > SARSA	Q-Learning performs better	Q-Learning is better under given data
4	Reward increases with γ	Long-term matter	rewards $\gamma=0.9$
5	Success $45\% \rightarrow 92\%$	Major improvement	policy Strong learning, optimality not proven
6	0.05 most stable	Balanced updates	LR=0.05
7	ϵ -greedy highest reward	Effective exploitation	ϵ-greedy
8	$50 \rightarrow 15$ steps	70% improvement	Efficient policy; optimality needs more evidence
9	$\gamma=0.95$ highest reward	Future rewards important	$\gamma=0.95$
10	Rewards stabilize after 500	Policy becomes consistent	Evidence of convergence/robustness